IoT-Based Smart Robotic Assistance for Alzheimer's Patients

Intelligent Healthcare Monitoring & Assistance System

Abstract:

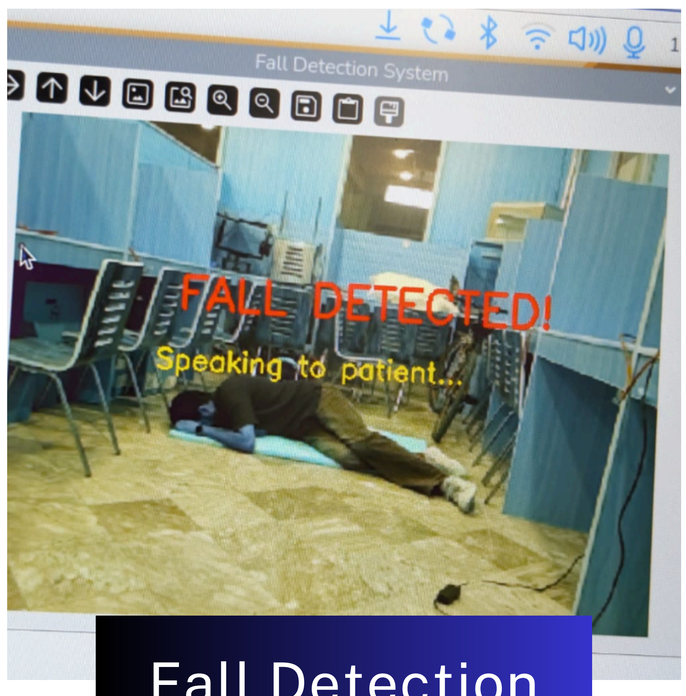
AThe IoT-Based Smart Robotic Assistance System for Alzheimer's Patients is an AI-powered healthcare robot designed to enhance patient safety, monitoring, and caregiver support. The system utilizes computer vision and artificial intelligence to perform real-time fall detection, patient tracking, and family member recognition. It continuously monitors patient activity, detects emergencies, and sends instant alerts to caregivers through Firebase cloud integration and a dedicated mobile application. The solution also incorporates an intelligent medication reminder system that provides timely voice alerts to patients. By combining robotics, AI, IoT, and cloud technologies, the proposed system aims to improve the quality of life, independence, and healthcare management of Alzheimer's patients while reducing caregiver burden.



Objectives:

- To detect realtime fall using YOLOv8n-pose and computer vision algorithms.
- To autonomously track/follow patient with safe-distance robot navigation for continuous monitoring.
- To perform Family member recognition using facial recognition technology.
- To generate Voice-based medication reminders through scheduled alerts through App.
- To facilitate Real-time IoT monitoring of temperature, gas, and humidity sensors.
- To enable Instant emergency notifications to caregivers via Firebase and mobile application.
- To provide live streaming of patient remotely.

Empowering Independent Living Through Intelligent Robotic Healthcare Assistance.



Fall Detection

Detects falls and immediately sends alerts to caregivers through the mobile application.



Patient Tracking

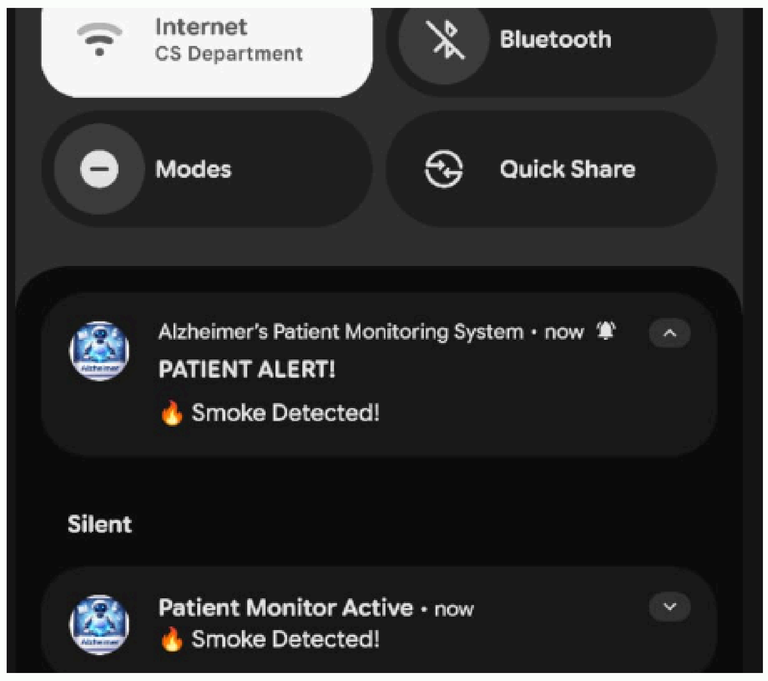
Autonomously follows the patient while maintaining a safe distance using intelligent motor control.



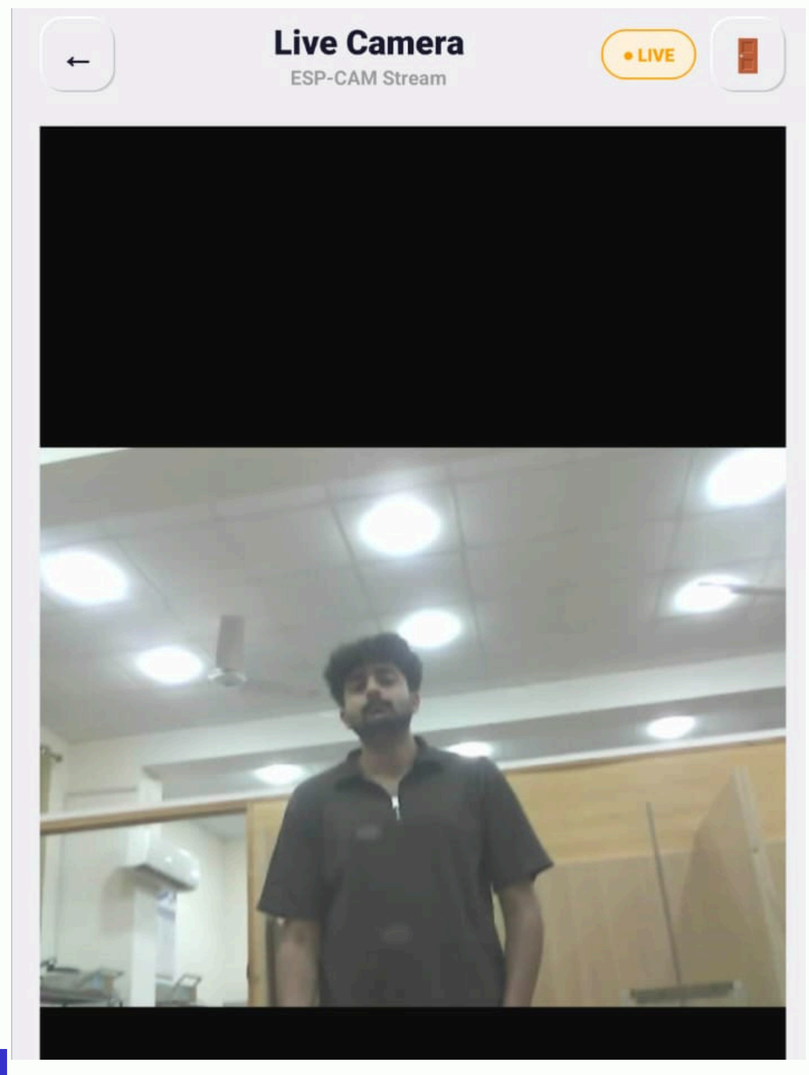
Family Recognition

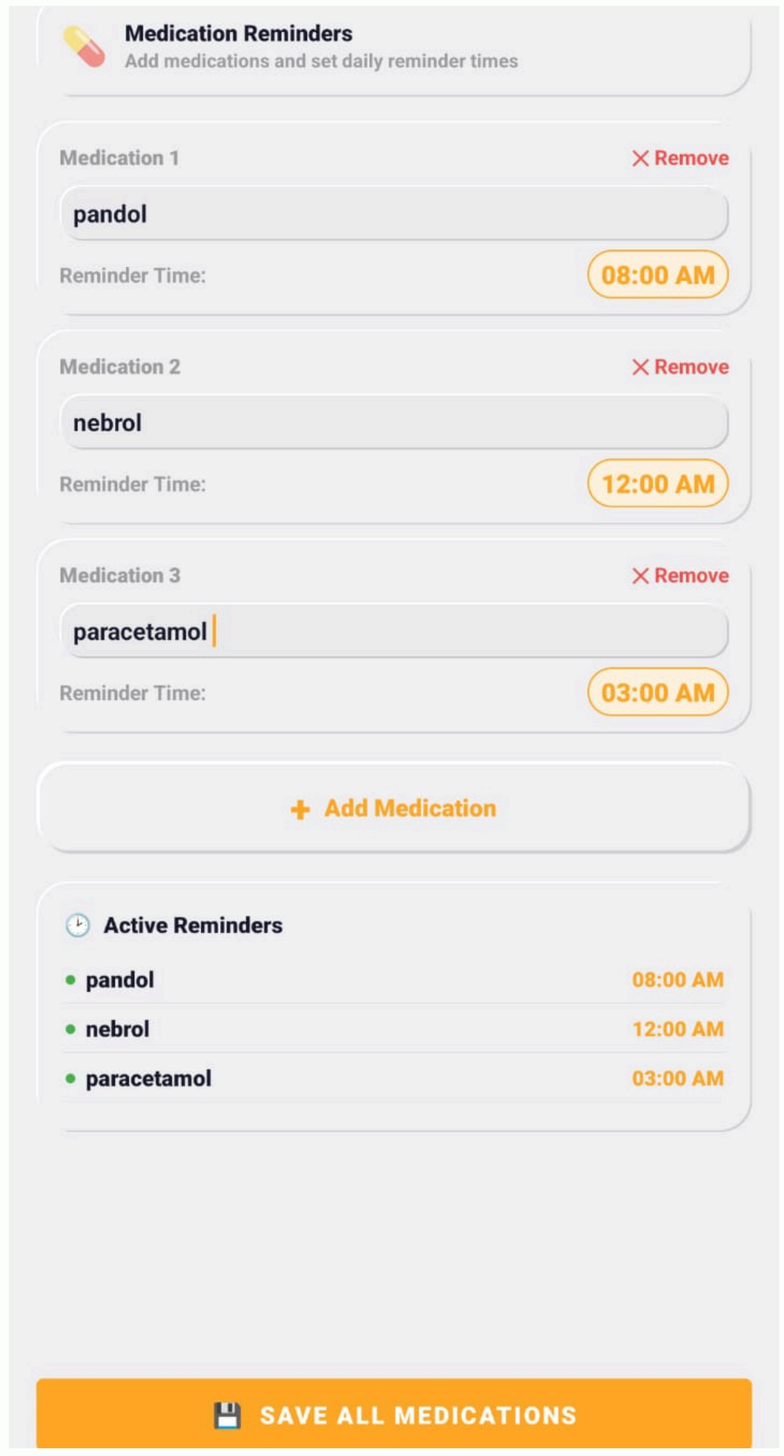
Recognizes registered family members using facial recognition and provides personalized interaction.

Smart Healthcare Through Artificial Intelligence & IoT



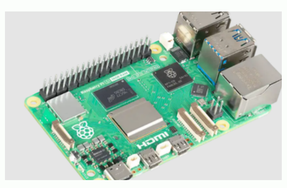
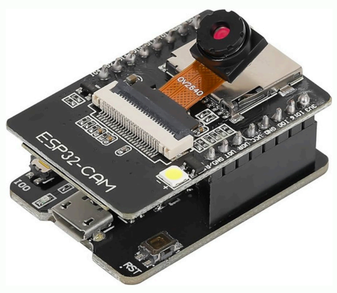
Real-time monitoring as live footage of patient,, sensor readings, alerts, and medication schedules.

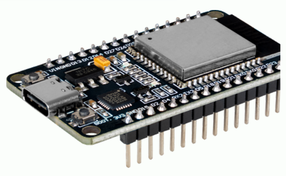
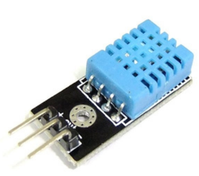




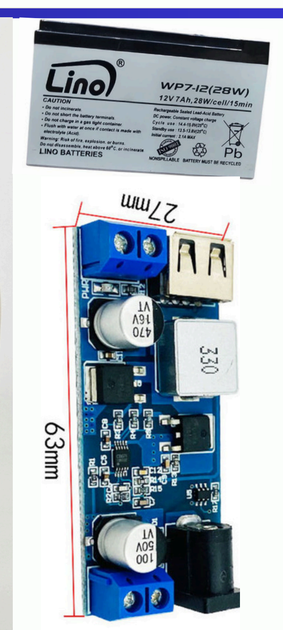
Secure communication between robot, caregivers, and mobile application .Continuous tracking, fall detection, and emergency notification system.

Hardware Components Used










Technology Used

- Raspberry Pi 5
- Pi Camera v3.1
- YOLOv8
- Firebase
- OpenCV
- Facial Recognition Model & classifiers
- Android Mobile Application
- IoT Sensors

Key Features

- Fall Detection
- Patient Tracking/following
- Family Member Recognition
- Medication Reminders
- Mobile App Integration
- Firebase Cloud Integration
- Temperature Monitoring
- Gas Detection
- Smoke Detection
- Emergency Alerts

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